

Java Programming Syllabus



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Java Full stack Development Course

Content

Introduction

1. What is Front End Development
2. Front End Technologies
3. What is Back End Development
4. Backend Development Technologies
5. What is Full stack Development
6. Advantages of Full stack Development
7. Roles of Full stack Developer

Module-1: Backend Development

Core Java

8. What is Programming Language?
9. Types of programming languages
10. Java Introduction
11. History of Java
12. Principles of Java
13. Java Installation
14. Java Program Execution Flow





15. JDK vs JRE
16. JVM Architecture
17. Identifiers & Keywords
18. Data Types
19. Variables
20. Methods
21. Arrays
22. Strings Handling
23. Object Oriented Principles
24. Constructors
25. Interfaces
26. Programs related to OOPs
27. Final classes
28. Abstract Class
29. Exception handling
30. Multi-Threading
31. Concurrency Package
32. Executor Framework
33. Synchronization
34. Lock mechanism
35. Serialization
36. Wrapper classes





37. Auto Boxing, Un-Boxing
38. Java Memory Management
39. Class Loader Implementation
40. Variable Arguments
41. File IO Operations
42. Introduction to Collection Framework
43. List
44. Set
45. Map
46. Properties
47. Internal implementation of collection framework
(List, Map, Set etc)
48. Java 8 Features Introduction
49. Lambda Expressions
50. Functional Interfaces
51. Default & Static Methods
52. Method References
53. Date and Time API changes
54. Stream API
55. Optional class
56. Spliterator Advanced Java
57. Introduction to Advanced Java





- 58. Introduction to Database
- 59. Types of databases
- 60. Database Environment Setup
- 61. DB Table Creation
- 62. Table Altering
- 63. CRUD Operations
- 64. Primary Keys
- 65. Foreign Keys
- 66. JDBC Introduction
- 67. JDBC API Details
- 68. JDBC Architecture
- 69. JDBC Driver
- 70. Driver Manager
- 71. Connection
- 72. Statement
- 73. Prepared Statement
- 74. Callable Statement
- 75. Result Set
- 76. What is Dao Layer?
- 77. Steps to create JDBC Application
- 78. JDBC Applications Development
- 79. CRUD Operations





80. Connection Pooling
81. Most common exceptions in JDBC apps
82. Servlets
83. What is web application?
84. Web application architecture
85. Client & Server
86. Http Protocol
87. What is Servlet?
88. Servlet API
89. Servlet Life Cycle
90. Steps to create Servlet
91. Deployment Descriptor
92. Servlet interface
93. Generic Servlet class
94. HttpServlet class
95. HttpServletRequest
96. HttpServletResponse
97. ServletConfig
98. ServletContext
99. Init Params
100. Context Params
101. Forms development





- 102. Cookies
- 103. Hidden variables
- 104. Session Tracking in Servlets
- 105. JSP
- 106. Introduction to User interface
- 107. Introduction to JSP
- 108. Servlets vs. JSP
- 109. Life Cycle of JSP
- 110. JSP Implicit Objects
- 111. JSP Directives
- 112. JSP Scripting Elements
- 113. MVC Architecture Introduction
- 114. Forms Development
- 115. Mini Projects Development

Module-2: Web Development

HTML

- 1. Web Development Overview
- 2. HTML Introduction
- 3. HTML Document Structure
- 4. Headings
- 5. Paragraphs
- 6. Typography





7. HTML Links
8. Images in HTML
9. Attributes
10. Lists
11. HTML Tables
12. Forms & Inputs
13. Block & Inline elements
14. Divs
15. Spans
16. Classes & Ids
17. Iframes
18. HTML 5 Semantics CSS
19. CSS Introduction
20. Basic CSS Selectors
21. Dev Tools Introduction
22. Fonts
23. Colors
24. Backgrounds
25. Borders
26. Box Model
27. Margin & Padding
28. Display, Alignment & Positioning





- 29. Pseudo Selectors
- 30. Responsive Layouts
- 31. Introduction to FlexBox
- 32. Building Responsive Website
- BOOTSTRAP
- 33. Bootstrap Introduction
- 34. Typography & Utility Classes
- 35. Buttons & Button Groups
- 36. NavBars & Navs
- 37. List Groups
- 38. Forms & Inputs
- 39. Input Groups
- 40. Tables
- 41. Pagination
- 42. Alerts
- 43. Spinners
- 44. Cards
- 45. Carousel Slider
- 46. Collapse
- 47. Accordion
- 48. Tooltips
- 49. Popovers





50. Models

JAVA SCRIPT

51. Introduction to Java Script

52. Script Tags

53. Console Output

54. Variables

55. Data Types

56. Strings & Strings Methods

57. Arrays

58. Java Script Objects

59. JSON

60. Arrays

61. Loops

62. Array Helpers

63. Conditionals

64. Functions

65. Constructor Functions

66. ES6 Classes

67. Window Object & DOM

68. DOM Selection

69. DOM Manipulation

9.





70. Events

71. Handling Forms

Front End Projects Development

72. Building App with HTML & CSS

73. Building App with HTML, CSS & Bootstrap

74. Building App with HTML, CSS, Bootstrap & Java Script

75. Resume Preparation for Frontend Development

76. Roles & Responsibilities of Frontend Developer

77. Front end Interview Questions & Answers

78. Interview Tips & Tricks

SPRING BOOT & MICRO SERVICES COURSE CONTENT

SPRING BOOT

1) What is Spring Framework

2) What is Spring Boot

3) Differences between Spring & Spring Boot

4) Spring Boot Overview

5) Pros & Cons of Spring Boot





6) Approaches to create Spring Boot Application

a) Spring Initializer (start.spring.io)

b) Spring Starter Wizard in STS IDE

7) Introduction to Spring Boot Starters

a) Spring Boot Parent Starter

b) Spring-boot-starter

c) Spring-boot-starter-web

d) Spring-boot-starter-webflux

e) Spring-boot-starter-data-jpa

f) Spring-boot-devtools

g) Spring-boot-starter-mail

h) Spring-boot-actuator

i) Spring-boot-starter-test etc.

8) What is Start Class in Spring Boot

9) @SpringBootApplication annotation internals

10) SpringApplication.run(..) method internals

11) Spring Boot Application Boot strapping

12) AutoConfiguration in Spring Boot

13) IOC container

14) Dependency Injection a) Setter Injection

b) Constructor Injection c) Field Injection





15) Stereotype Annotations

a) @Component

b) @Service

c) @Repository

d) @Controller

16) Base Package Naming convention

17) Component Scanning

18) Auto wiring

a) byType

b) byname

c) constructor

d) none

19) @Qualifier annotation

20) @Primary annotation

21) Introduction to @Configuration annotation

22) Details of @Bean annotation

23) Bean Life Cycle

24) Bean Scopes

25) Banner in Spring Boot

26) Standalone application development in Spring Boot





27) Standalone application with layered architecture

28) Runner in Spring Boot

a) Application Runner

b) Command Line Runner

SPRING DATA JPA

1) What is Persistence Layer

2) Best practises to follow in persistence layer

3) ORM Basics

4) Spring Data JPA Introduction

5) Differences between Spring ORM and Spring Data

6) CrudRepository introduction

7) CrudRepository methods for DB operations

a) save() method

b) saveAll() method

c) findById() method

d) findAllById() method

e) findAll() method

f) deleteById() method

g) deleteAllById() method

h) delete() method i) count()

j) existsById()





- 8) Custom findByXXX method syntax
- 9) Custom Queries Execution in Data JPA
- 10) JpaRepository introduction
- 11) JpaRepository methods for DB operations
- 12) Pagination Using Data JPA methods
- 13) Sorting Using Data JPA Methods
- 14) Query by Example Executor
- 15) Generators
- 16) Custom Generators in Spring Data
- 17) Embedded Database Introduction
- 18) Application Development using Embedded Database
- (H2)
- 19) Application Development Using MYSQL Database
- 20) Profiles in Spring Boot SPRING WEB MVC
- 21) Spring Web MVC Introduction
- 22) Spring Web MVC Advantages
- 23) Spring MVC Architecture
- 24) Introduction to Front Controller
- 25) Controllers
- 26) Handler Mappers
- 27) View Resolvers
- 28) Web Application development using Spring Boot





29) Embedded HTTP Servers Introduction

a) Embedded Tomcat Server

b) Embedded Jetty Server

30) Making Jetty as Default server

31) Web Application Deployment in External Server

32) Sending Data From UI to Controller

a) Query Param

b) Path Param

33) Sending Data From Controller to UI

a) Model

b) ModelAndView

34) @RequestBody annotation

35) @ResponseBody annotation

36) Form Based application development using Spring Boot

37) Thymeleaf Introduction

38) Web Application with Thymeleaf

39) Sending Email using Spring Boot

40) Exception Handling in Spring Boot Web Application

41) Spring Boot Actuators





- a) Health
- b) Info
- c) Heapdump
- d) Theaddump
- e) Beans
- f) Httptrace
- g) Mappings
- h) Shutdown etc

SPRING REST

- 42) Distributed Applications
- 43) Distributed Technologies
- 44) SOAP vs REST
- 45) RESTful Services Introduction
- 46) REST principles
- 47) One Time operations
- 48) Run Time Operations
 - a) Marshalling
 - b) Un Marshalling
- 49) JAX-B Introduction
- 50) JAX-B Architecture





- 51) Applications development with JAX-B
- 52) JSON Introduction
- 53) XML vs JSON
- 54) JACKSON API
- 55) Converting Java object to JSON and vice versa using Jackson API
- 56) GSON API
- 57) Converting Java Object to JSON and Vice Versa using GSON API
- 58) HTTP Protocol Details
- 59) HTTP Methods
 - a) GET
 - b) POST
 - c) PUT
 - d) DELETE
- 60) HTTP Status Codes
- 61) @RestController
- 62) @RequestBody
- 63) @ResponseBody
- 64) @RequestParam
- 65) @PathVariable





- 66) MediaTypes
- 67) Consumes
- 68) Produces
- 69) Accept Header
- 70) Content-Type header
- 71) REST API Development using Spring Boot
- 72) POSTMAN
- 73) SWAGGER & SWAGGER UI
- 74) Exception Handling in REST API
- 75) REST Security
 - a) HTTP Basic Auth
 - b) JWT
 - c) OAuth2.0
- 76) Mono Objects
- 77) Flux Objects
- 78) REST Client Introduction
- 79) RestTemplate
- 80) WebClient
- 81) RestTemplate vs WebClient
- 82) Reactive Programming
- 83) Synchronous vs Asynchronous Calls





- 84) Apache Kafka with Spring Boot
- 85) Redis Cache Integration with Spring Boot MICRO SERVICES
- 86) Monolith Architecture Introduction
- 87) Monolith Architecture case study
- 88) Monolith Application Deployment Process
- 89) Load balancer (Cluster) case study
- 90) Load Balancing Algorithms
 - a) Round Robin
 - b) IP Hashing
 - c) Sticky Session
- 91) Monolith Architecture Drawbacks
- 92) Micro services Introduction
- 93) Micro Services Advantages
- 94) Micro Services Dis-Advantages
- 95) Micro Services case study
- 96) Identifying Micro services boundaries 97) Micro services Architecture
- 98) Micro services Development
 - a) API – 1
 - b) API – 2





- 99) Interservice communication case Study (FeignClient)
- 100) Cloud Introduction
- 101) AWS Account Creation
- 102) AWS Services Overview
- 103) Deploying Micro services to AWS
- 104) Service Registry case study (Netflix Eureka)
- 105) API Gateway
- 106) Hystrix (Circuit Breaker)
- 107) Spring Boot Admin Server
- 108) Spring Boot Admin Client
- 109) Distributed Logging
 - a) Sleuth Logging
 - b) Zipkin Server
- 110) Ribbon case study (Micro services load balancer)
- 111) Implementing Ribbon in Micro services
- 112) ConfigServer Introduction & Implementation
- 113) Micro services Integration with Angular TOOLS
- 114) Maven
- 115) Log4J
- 116) Junit 5
- 117) Jenkins Deployment
- 118) Docker





119) POSTMAN

120) Swagger

JAVA REAL-TIME PROJECT COURSES

CONTENT

INTRODUCTION

- 1) Introduction to Real-time Project
- 2) Types of Software companies
 - a) Product Based Companies
 - b) Service Based Companies
 - c) Outsourcing Companies
- 3) Interview process in software companies
- 4) Types of software jobs
 - a) Permanent Job
 - b) Contract Job
- 5) Types of software projects
 - a) Scratch Development Projects
 - b) Maintenance Projects
 - c) Migration Projects





6) Types of teams in software project

a) On-shore Team

b) Off-shore Team

7) How to collaborate with team members

8) Discussion related to Bridge calls & Official emails

9) Role Chart Details REALTIME – TOOLS

10) Maven

11) Gradle

12) Debugging

13) Project Lombok

14) GIT

15) BitBucket

16) Log4J

17) SLF4J

18) Putty & WinScp

19) Splunk

20) JUNIT

21) Mocking

22) JaCoCo

23) SonarQube





24) JIRA

25) Jenkins

26) Dockers

27) Kubernetes

28) JMETER

29) Apache Kafka

30) Redis Cache

31) POSTMAN MINI PROJECTS DEVELOPMENT

32) SDLC (Agile)

I. Story

II. Story Points

III. Backlog grooming

IV. Sprint Planning

V. Sprint

VI. Sprint Duration

VII. Scrum

VIII. Mid-iteration review

IX. Retrospective

33) Real-time Environment Terminology

34) Requirements Gathering

35) Minutes of meeting





- 36) Query Log
- 37) DB Design
- 38) DB Template
- 39) Real-time Coding Standards
- 40) 01-Mini Project
- 41) 02- Mini Project
- 42) 03- Mini Project
- 43) Mini Projects Case study MICROSERVICES
- 44) Monolith Vs Microservices
- 45) Microservices Advantages
- 46) Microservices Architecture
- 47) Service Registry (Eureka)
- 48) API Gateway (Cloud Gateway)
- 49) Gateway Routing
- 50) Gateway Filters
- 51) FeginClient
- 52) Microservices Case Study MAJOR PROJECT
DEVELOPMENT
- 53) Major Project Introduction
- 54) Major Project Modules
- 55) Major Project Architecture
- 56) Major Project Functional Flow





- 57) Database Design
- 58) Story Creation in JIRA
- 59) Story Development
- 60) Logging in Project
- 61) Swagger Configuration
- 62) Actuators Configuration
- 63) Exception Handling
- 64) Multi-Threading use case in Project
- 65) Rest Calls in Project
- 66) Batch Processing
- 67) Unit Testing
- 68) Integration Testing
- 69) Code Review
- 70) Project Deployment AWS (Cloud)
- 71) What is Cloud Computing
- 72) AWS Introduction
- 73) AWS Advantages
- 74) AWS Account Creation
- 75) AWS Services Walk through
- 76) Launching EC2 Machine (Linux)
- 77) Connecting to EC2 using Putty
- 78) Linux Commands





- 79) S3 Buckets
- 80) RDS (Relation Database System)
- 81) Deploying Spring boot app in AWS
- 82) Load Balancing
- 83) Auto Scaling
- 84) IAM User ANGULAR FRAMEWORK
- 85) Angular Introduction
- 86) Angular JS vs Angular
- 87) Angular Version History
- 88) Angular Environment Setup
- 89) Angular Architecture
- 90) Angular App Development
- 91) Components
- 92) Templates
- 93) Directives & Pipes
- 94) Services
- 95) Forms Development
- 96) Form Validations
- 97) Routing
- 98) Http Client Module
- 99) Integration (Front End + Back End)
- 100) Spring Boot with Angular Integration





- 101) Angular App Deployment
- 102) Heroku Deployment INTERVIEW GUIDE
- 103) Resume Preparation
- 104) Profile Creation in job portals
- 105) Email Writing
- 106) HR Discussion details
- 107) Joining Formalities
- 108) First day and last day in office
- 109) Daily Activities in Job
- 110) Interview Tips & Tricks

